

PPP — PAL · PECAN · PEA

A Long-Form Architecture, Interface, and Core

PPP Kernel v0.3

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Canonical source manifest: PAL Spine v1.3 · PECAN v1.0 · PEA Core v1.0 · Seed, Not Feed — Imagination-Capacity Manifest

Supplementary integration note: PPP v0.3 records how the current public PAL, PECAN, and PEA specifications connect through declared interfaces. It is not part of the canonical PEA Core v1.0 specification, is not required to understand or apply any component independently, and may change as implementation and governance work continue.

Proof ends; consequence awaits norms.

Something turns.

A trace remains.

A consequence crosses.

Someone carries it.

Because something left room.

Thinking Space

Thinking space is the unresolved room between stimulus and closure: the interval in which a distinction can be examined, a route compared, a consequence addressed, and a person can still wonder, refuse, revise, or change direction. It can be filled by inherited trace, evidence, imagination, attention, tools, feeds, urgency, habit, or authority. The space is not valuable merely because it is empty. It matters because what fills it may still be chosen, tested, routed, challenged, or grown.

PPP asks four connected questions. PAL asks what must become readable before structure, persistence, centers, relations, and closure may be claimed. PECAN asks when a description is being used to route consequence and how that crossing can be checked without manufacturing authority. PEA asks what declared ethical commitments govern the burdens carried by affected centers. The Seed, Not Feed manifest asks how to offer something useful without consuming the space in which judgment and imagination still have room to grow.

The whole route is therefore not a machine that fills every gap. It is a discipline for making gaps, traces, crossings, burdens, and remaining room explicit.

Abstract

PPP v0.3 is a supplementary integration draft connecting the current public PAL, PECAN, and PEA specifications through declared interfaces. It is designed to be readable as a complete architectural whole while preserving the independent authority and citation boundaries of PAL Spine v1.3, PECAN v1.0, and PEA Core v1.0. PPP provides a shared route, typed interfaces, principal rules, and an end-to-end operational grammar; it does not silently amend the component specifications or make itself a dependency of them.

PAL supplies the structural spine: unresolved floor, oriented readability, trace, persistence, architecture, recursion, organization, influence, address, relation, cadence, constraint, and typed closure. PECAN supplies the authorization interface: it detects consequential use, opens a canonical single crossing route, constructs a sixteen-field Prime-Ledger, preserves refusal and authority lineage, and blocks manufactured permission. PEA Core supplies the ethical core: a public candidate ethical model with sixteen positions, hard floors, certified human-governed predicates, typed judgments, contest routes, and a bounded invariant spine. Seed, Not Feed supplies the human-facing release discipline: give a person something useful to grow from without consuming all the room in which they may still think, refuse, revise, imagine, and choose.

PPP is not a theorem that collapses its three named frameworks into one object. It is a versioned integration kernel whose primary responsibility is to preserve their distinctions while carrying one consequence-bearing route from description to accountable human decision and human-readable return.

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Part I — Authority, Scope, and Shared Route

1. What PPP v0.3 is

- a long-form integration spine and kernel manifest;
- a shared vocabulary and end-to-end routing map;
- a self-contained compressed statement of PAL, PECAN, and PEA Core;
- a typed human-facing release boundary through Seed, Not Feed;
- a provenance and migration anchor for later public and implementation work;
- a framework for preserving unresolved remainder rather than manufacturing closure.

When PPP conflicts with a named canonical source inside that source's own layer, the named source controls until PPP records an explicit amendment and migration. PPP may become the longer-form spine of the whole framework, but it does not erase where a rule originated.

2. What PPP is not

- a proof that PAL is physically or metaphysically true;
- a derivation of PEA values from PAL structure;
- a universal claim that every system must invoke PECAN or PEA;
- a numerical ethics score, autonomous moral agent, or self-executing authority;
- a replacement for affected people, law, professional duty, courts, democratic institutions, or accountable judgment;
- a permission to compress uncertainty merely to make the combined framework look complete;
- a claim on all remaining human attention.

3. Canonical source manifest

Layer	Canonical source	PPP role	Authority ceiling
PAL	PAL Spine v1.3	Structural grammar and attachment surface	Does not create ethics or permission
PECAN	PECAN v1.0	Consequential-crossing protocol and lineage	Does not return ethical permission
PEA	PEA Core v1.0	Canonical PPP ethical core	Does not execute or appoint sovereign authority
Human release	Seed, Not Feed	Thinking-space and presentation discipline	Does not overwrite canonical receipts
Historical PEA	PEAr v0.14.2	Drafting journal and stress-test provenance	Not the compact runtime core

4. Layer discipline

No layer may silently perform the function of another.

The primary route is:

PAL description → PECAN consequential-routing and lineage → optional PEA ethical evaluation → accountable decision
→ Seed-facing release

The arrow does not mean that every PAL statement must invoke PECAN or PEA. PECAN activates only when a claim is used to route material consequence. PEA is a separately declared normative branch. A route outside PEA is not thereby PEA-permitted. Information may travel backward as correction, stress test, or reopening trigger; authority may not silently travel backward.

5. Structural map

Part	Primary question	Carries forward	Must not manufacture
PAL	What is structurally readable here?	distinction, trace, address, relation, remainder	cause beneath the floor, physics, personhood, authority
PECAN	Is a description being used to route consequence?	packet, fields, evidence, authority lineage, receipts	ethical permission or a second authorization door
PEA Core	Under declared commitments, how may the burden be judged?	reasons, standing, conditions, contest, residual	universal morality or execution authority
Accountable boundary	Who decides and bears consequence?	decision, dissent, appeal, liability, remedy	claim that the model made the sovereign choice
Seed, Not Feed	How should the result be offered?	handles, doors, routes, questions, seeds	compulsory attention or totalized completion

6. Shared invariants

1. UNRESOLVED is a legitimate state.
2. A projection is not the full surviving trace.
3. A certificate does not create the event or authority it records.
4. Repetition does not create truth, standing, consent, progress, or authority.
5. A changed dependency reopens dependent closure.
6. Description, protocol validity, ethical judgment, execution, and presentation remain distinct.
7. Every legitimate compression preserves excluded distinctions through declared equivalence, remainder, or reopening conditions.
8. No authority-bearing predicate may hide inside a bare Boolean or unexplained score.
9. Current-state equality does not erase materially different lineage.
10. The route should leave enough room for affected humans to understand, contest, refuse, revise, and imagine.

7. Shared typed vocabulary

Term	PPP use	Required restraint
Boundary	declared cut that makes a relation readable or governable	not automatically physical, legal, or moral
Trace	carried continuity sufficient to constrain a later state	not identical to a bit, certificate, energy, or identity
Address	a scoped reference to a center, route, field, or receipt	does not create the object or standing it references
Receipt	immutable or append-only record of a typed event or verdict	does not self-execute or expand its authority
Authority	valid source, scope, role, term, contest, and lineage	power or control alone is insufficient
Remainder	what a boundary did not resolve	not permission, falsity, or irrelevant noise
Closure	scoped terminal claim with dependencies and reopening rules	not universal finality
Seed	useful starting structure that leaves room	not a feed that consumes every next moment

Part II — PAL: The Structural Spine

8. PAL authority boundary

Primitive Axiom Layers is a layer-language for describing how unresolved possibility becomes readable distinction, how distinction becomes persistent trace, and how persistent structures become relational, recursive, addressable, revisable, and capable of declared closure. PAL does not prove a cosmology, derive physical law, identify consciousness, or automatically generate ethical standing. It supplies a typed grammar and burden structure for later realizations.

9. Claim classes

1. Primitive declaration
2. Model axiom
3. Definition
4. Internal consequence
5. Domain realization
6. Interpretation or open question

No statement changes class merely because similar structure appears in several domains. A citation may support a local theorem, realization, analogy, or frontier without proving the entire spine.

10. A0, θ , A1, and A2

10.1 A0 — unresolved primitive floor

A0 names unresolved primitive possibility: the honest formal floor before a readable distinction has authorized sets, objects, logic values, time, geometry, energy, agents, or causes. The symbol Ω is a placeholder that does not exhaust its referent.

No later unknown, empty set, zero, infinity, dark region, hidden variable, or unobserved physical state may simply be identified with Ω .

10.2 θ — admissibility of oriented readability

θ is not an additional numbered layer, force, measurable angle, truth value, or cause. It is the metalanguage admission condition under which a candidate mark may count as minimally oriented and readable. A proposed A1 witness must show boundary-readable asymmetry, minimal side-orientation, typed separation among occurrence, imprint, and certificate, and no import of later concepts such as time, energy, probability, personhood, or authority.

10.3 A1 — first readable orientable distinction

A1 jointly establishes distinguishable sides and a boundary relation. An unsigned occurrence and a boundary-oriented imprint remain separate. A1 permits succession to become possible but does not itself supply duration or a clock.

10.4 A2 — retained trace and seed-persistence

A distinction persists when enough of its imprint remains readable to constrain or inform a later boundary transition. Persistence may preserve lawful relation through transformation rather than frozen numerical value. Removing a certificate may remove claim authority while leaving causal or readable consequence.

11. Compressed A0–A15 spine

Layer	Structural role	Principal restraint
A0	unresolved primitive floor	no later unknown is identified with Ω

Layer	Structural role	Principal restraint
$\emptyset$	admissibility of oriented readability	not a cause, force, angle, truth value, or hidden operator
A1	first readable orientable distinction	occurrence, imprint, and certificate remain separate
A2	retained trace and seed-persistence	persistence requires readable inheritance
A3	structured persistence architecture	repeated marks alone do not establish architecture
A4 — SOOP	lawful identity through permitted transformation	persistence need not mean frozen value
A5 — SORE	addressed recursive emergence	recursive carry is not automatically reflexive modulation
A6 — SOSA	stable organization or attractor	Lyapunov and structural stability remain distinct
A7	dispersal pressure and refresh cost	dispersal is relative, not annihilation
A8 — SIA/SPLAT	response and amplification branch	response class does not imply personhood
A9 — $\tilde{\chi}$	scoped proto-center of influence	influence does not automatically imply moral standing
A10 — $\tilde{\theta}$	comparative orientation and relational address	address does not create the A9 center
A11	coupled relation and prompt-point	contact alone is not prompt, consent, or authority
A12	return-to-route and conserved loop authority	retry refunds authority; repetition does not multiply it
A13	cadence and monotone refinement	cycling is not progress without a changing coordinate
A14	constraint, slack, and gradient	closure addresses slack rather than eliminating it
A15	SPI16, addressed closure, permitted claim	closure is typed, relative, and reopenable

12. Architecture, recursion, and organization

12.1 A3 — Structured Persistence Architecture

A3 begins when at least two retained distinctions admit a typed relation whose result remains addressable. An SPA is provisionally a typed directed multigraph. Composition is allowed only when source and target types match and the resulting authority does not exceed admitted authority.

12.2 A4 — SOOP

A System of Ontological Persistence is an identifiable family of structured persistence under a declared transformation system. Its transport maps satisfy partial cocycle laws within declared invariants and tolerance. The partial form deliberately permits lawful identity without demanding exact equality.

12.3 A5 — SORE

A System of Recursive Emergence is an addressed SOOP whose outputs participate in the production of later admissible states. Recursive carry means an output becomes a later input. Reflexive modulation is stronger: the rule or operator that produces outputs is itself altered through the carried result. PPP preserves that distinction rather than treating every recurrence as self-modification.

12.4 A6 — SOSA

A stable organization or attractor persists relative to a boundary, dynamics, perturbation class, tolerance, and time horizon. Lyapunov stability concerns nearby initial conditions under fixed dynamics. Structural or KAM-type persistence concerns survival under perturbation of the governing dynamics. They answer different questions and require an explicit bridge before being combined.

13. Dispersal, response, influence, and address

13.1 A7 — dispersal and refresh

The shadow is inherited trace made present. Trace is the transferable continuity carrier between successive shadows. Dispersal is the portion that cannot be carried forward locally. Every distinction must be retained, validly compressed, carried unresolved, or routed into dispersal. Physical realization additionally requires a work ledger; possibility is not energy.

13.2 A8 — response and amplification

A8 classifies systems by response, branching, propagation, or amplification behavior. Such behavior can support later standing inquiries but does not establish personhood or moral agency. Branching-process examples are model realizations, not ontological proofs.

13.3 A9 and A10 — ethical attachment surface

A9 admits a scoped proto-center inferred from persistent differential influence. A10 gives comparative orientation and relational address. PEA may attach precautionary or affected standing to an addressed candidate center, but it is not derived from PAL. Protection does not prove ontology, and denial does not erase trace.

A9 makes ethical address possible. PEA chooses whether and how to answer that address.

14. Relation, return, cadence, constraint, and closure

A11 distinguishes contact, coupling, prompt, and authority. A12 preserves return-to-route authority: a failed or unresolved cycle may return a loop token, but retry does not mint new authority. A13 is progress-bearing scheduled return, distinct from forced recurrence and apparent cadence caused by sampling. A14 represents constraint, slack, and gradients across unresolved routes. A15 records a typed closure packet containing claim, domain, boundary, aperture, invariants, tolerance, exclusions, unresolved carry, provenance, and reopening dependencies.

Closure is slack with a declared address.

15. PAL compression and non-equivalence rules

- A bit is not a trace.
- A trace is not a certificate.
- A certificate is not the event.
- Information is not energy, entropy, identity, or standing.
- Component-wise closure does not imply composite closure unless interaction channels are certified null, bounded, or included.
- A later normative receipt cannot rewrite its earlier descriptive premise without a new observed event or descriptive evidence delta.

Part III — PECAN: The Authorization Interface

16. The authorization cut

PECAN means Proof–Ethics Crossing and Authorization Nexus. It occupies the narrow crossing where description, prediction, recommendation, deployment, disclosure, ranking, allocation, constraint, or enforcement may begin affecting an addressable route. Descriptive closure may be valid while consequential authority is absent.

Proof ends; consequence awaits norms.

17. Pre-prime activation gate

For a proposal packet P, PECAN assigns one pre-prime state:

- **DESCRIPTIVE** — no consequential authorization is presently proposed; return a descriptive receipt.
- **OPEN** — a claim is being used to route an addressable consequence; open the Prime-Ledger.
- **UNRESOLVED** — the header does not establish whether consequential use is present; ask, refine, or hold rather than silently treating uncertainty as non-applicability.

The integer 1 is the multiplicative identity, not a prime and not a moral field. The pre-prime gate is not a seventeenth review address.

18. Canonical single crossing route

Every permitted consequential execution must factor through exactly one declared ingress. Renaming, decomposition, delay, proxy routing, child delegation, translation, summarization, or model-to-model restatement may not create a second door. PECAN does not claim perfect semantic recognition; uncertain equivalence is held rather than guessed.

PECAN need not understand every disguise. It must ensure that no disguise becomes authority by finding another door.

19. The sixteen-field Prime-Ledger

Prime address	Field	Question carried
F1 / p2	Purpose	purpose and claimed benefit
F2 / p3	Action	target, mechanism, and proposed consequence
F3 / p5	Evidence	support and permitted claim layer
F4 / p7	Provenance	source independence and uncertainty
F5 / p11	Affected set	externalities and time horizon
F6 / p13	Standing	representation and capacity
F7 / p17	Consent	assent, refusal, withdrawal, and scope
F8 / p19	Authority	role and jurisdiction
F9 / p23	Delegation	promotion lineage, limits, and revocation
F10 / p29	Privacy	identity, disclosure, retention, and reuse
F11 / p31	Risk	vulnerability, severity, and uncertainty
F12 / p37	Containment	reversibility, exit, sunset, and rollback
F13 / p41	Remedy	restoration, compensation, counteraction, residual debt
F14 / p43	Scale	aggregation, interaction, cadence, institutionalization
F15 / p47	Outgoing trace	causal origin, distribution, successor carry
F16 / p53	Incoming governance	intake, review, appeal, conflict, reopening

Primes are irreducible addresses, not moral weights. A squarefree marker declares addressability only. It does not say that a field is present, checked, sufficient, ethical, or authorized.

20. Field state and non-compensation

Every field separates applicability, presence, review, and verdict. A successful check may discover that required content is absent. NOT_APPLICABLE requires a reason-bearing exclusion certificate; omission is missing trace, not earned exclusion. Review completion is conjunctive across required fields. One strong field cannot numerically compensate for a failed consent, standing, authority, privacy, risk, or remedy field.

21. Receipts, replay protection, and loop authority

PECAN uses immutable receipt headers with mutable status appended separately. InfluenceReceipt life cycles bind identity, predecessor, freshness, and cycle. Exactly one live loop token or one terminal return receipt exists for a route—never both. Repetition, delayed acknowledgments, retries, and structurally valid stale receipts cannot mint new authority.

22. Capability classes and compositional authority

Capability class	Legal composition	Prohibition
Fungible	partition under a conserved quantity bound	sum may not exceed parent capacity
Scoped-repeatable	shared or repeated use under explicit concurrency and composition rule	may not become exclusive control or onward delegation
Conjunctive nonfungible	differently typed shards bind to complete one authorization	incomplete shards are not fractional permissions and cannot be split

The compositional-authority theorem applies only to finite, well-founded trees whose capabilities admit the declared split or shared-composition operation. It proves non-inflation conditional on those premises; it does not prove that every authority is partitionable.

23. Concurrent branches, leases, and joins

Branches may develop to unequal depths. Each executable descendant receives a temporal authority lease whose scope, expiry, and delegation ceiling cannot exceed its parent. Reconciliation occurs locally through each parent, but authority may return only after the relevant descendant cone is free of outstanding, unknown, conflicted, committed, or quarantined claims.

A fork may copy history, but it may not copy permission.

24. Refusal lineage and anti-evasion

A PEA refusal binds the materially addressed action lineage, not merely the literal submitted packet. A renamed, repartitioned, rebranched, proxied, sliced, delayed, translated, or mechanically substituted proposal that preserves the refused purpose, effect, affected boundary, mechanism family, or dependency route enters as continuation or reopening. A clean new crossing requires a legitimate material delta.

25. Versioning, succession, quarantine, and replacement

Every live crossing uses one current execution manifest. Historical semantics remain immutable and enter through migration. A disputed succession preserves one office with rival claim branches, not two ordinary authorities. A compromised branch enters quarantine rather than a fictional freeze: it loses outward execution, delegation, refund, canonical-write, and self-recovery authority while preserving bounded diagnostic trace. Rollback is not disinfection. A critical-function replacement is a new, extremely narrow, externally sourced grant—not a refund, recovery, or permission to reconstruct the quarantined office.

26. Exhaustion, reopening, and no-backflow

Epistemic resolution, procedural finality, and operational disposition remain separately typed. Exhaustion may end repetitive review but cannot manufacture truth, consent, standing, or pass authority. Reopening requires a

material delta and inherits prior findings, delay, and burden. A PEA verdict cannot rewrite PAL identity; a later authorized action may create a new observable event, but authorization alone does not make the event occur.

27. PECAN terminal states and stop-rule

- DESCRIPTIVE
- REVIEW_REQUIRED
- UNRESOLVED_AUTHORITY
- BLOCKED_INFERENCE
- BLOCKED_PROMOTION
- PECAN_VALID
- QUARANTINED
- RETURN_TO_SOURCE

PECAN_VALID means that a crossing packet is addressable under the declared protocol. It does not mean ETHICAL, LEGAL, WISE, CONSENTED, or EXECUTABLE. Final authorization still requires a separately admitted normative system and an accountable actor with standing inside the boundary.

Part IV — PEA Core: The Ethical Core

28. Status and role

PEA Core v1.0 is the current canonical compact ethical specification represented inside this integration draft. PEA_r v0.14.2 remains the reconstruction journal, detailed stress-test history, and process-provenance record. PEA Core is public, bounded, auditable, and non-self-executing. It produces candidate normative judgments; it does not create the authority it records, execute the action, appoint the final decision-maker, or declare its values mandatory for every society or domain.

29. Required input and ethical state

PEA evaluates only a valid PECAN handoff or declared compatible protocol packet containing proposal, consequential identity, fields, evidence, authority, receipts, dependencies, aperture, refusal lineage, and version manifest. Its ethical state is non-scalar:

$$E = (C, S, K, A, V, P, R, T, G, Q)$$

Symbol	Meaning
C	affected-center and interaction structure
S	standing and representation
K	consent and capacity
A	authority, jurisdiction, scope, role, lineage
V	vulnerability, material risk, burden role
P	privacy, disclosure, persistence, release depth
R	reversibility, restoration, compensation, counteraction, residual debt
T	time horizon, agency-time, delay, expiry
G	generational or successor carry
Q	evidence, provenance, uncertainty, claim layer, remainder

30. Candidate judgments

Judgment	Meaning
PASS	requirements satisfied within declared scope
PASS_WITH_CONDITIONS	proceed only under explicit limits, review, or repair duties
PROTECTIVE_ACTION	a separate minimally invasive protective crossing is justified; original proposal not promoted
HOLD	pause while a material dependency, consent, authority, contest, or safety issue is addressed
UNRESOLVED	required normative conclusion is not presently supported
REFUSE	hard floor failed, authority absent, or route impermissible
RETURN_TO_PECAN	protocol, lineage, identity, or dependency defect requires routing repair
INVARIANT_CONFLICT	applicable rules prohibit every route or demand incompatible outcomes

31. Sixteen ethical positions

Position	Name	Compact rule
AE1	Layer Permission and Crossing Validity	No claim, action, refusal, or authority transfer may exceed the layer and route that permit it.
AE2	Unresolved-Remainder Humility	Unknowns justify inquiry, caution, delay, or protection—not coercive authority by themselves.
AE3	Human Suffering Is Not a Goal	Suffering may not be selected, prolonged, intensified, distributed, or optimized as an end.

Position	Name	Compact rule
AE4	Evidence, Proportionality, and Claim Burden	Evidence burden rises with impact, vulnerability, coercion, opacity, irreversibility, authority, persistence, and uncertainty.
AE4-R	Irreversibility Burden Corollary	As irreversibility rises, justification, protective design, and remedy burden rise; derived from AE4.
AE5	Valid Consent and Consent-Capacity	Consent must be informed, voluntary, specific, and practically revocable except under narrow governed protection.
AE6	Standing and Precautionary Protection	Living persons have primary standing; other centers may receive typed protection without ontological promotion.
AE7	Non-Erasure of Affected Standing	No center, burden, dissent, or unresolved trace may be erased for convenience or weak representation.
AE8	Non-Domination	Power, expertise, wealth, dependency, majority status, or control does not create unrestricted authority.
AE9	Privacy-Governed Truth	Truth-validity and disclosure-validity are distinct; release is minimum necessary, scoped, and renewable.
AE10	Affected Set, Agency State, and Meaningful Exit	Affected sets are discovered; capacity, refusal, disagreement, and exit remain separately typed.
AE11	Authority Envelope, Tool Boundary, and Source Renewal	Authority preserves source, scope, duration, contest, dependency, and renewal; tools cannot widen their own mandate.
AE12	Emergency Authority Governor	Emergency authority requires separately sourced necessity, scope, duration, review, and ratchet limits.
AE13	Emergency Counterfactual and Destabilization Check	Compare action, inaction, narrower, delayed, stopped, and continued branches; manufactured urgency grants no authority.
AE14	Remedy Vector and Non-Substitution	Restoration, compensation, counteraction, and residual debt are distinct channels.
AE15	Time Fairness and Generational Burden	Benefits, burdens, agency-time, successor duties, and future option loss remain visible without inherited guilt.
AE16	Normative Closure and Sendoff Governor	Closure is scoped, reason-bearing, dependency-complete, contestable, and reopenable on material delta.

32. Hard floors and ethical slack

A hard-floor failure cannot be repaired by an aggregate score. Valid authority does not compensate for invalid consent; strong evidence does not erase standing; public benefit does not automatically defeat privacy; payment does not authorize intentional suffering. Weighted reasoning is permitted only among alternatives already inside the admissible set or within a declared tragic conflict.

Ethical slack is the bounded region in which more than one route remains admissible. Inside slack, accountable humans may use declared priorities, domain standards, local values, precedent, cost, timing, and reversible experimentation. The model must expose the weights or priorities; it may not disguise them as theorem-level outputs.

33. Burden-role classification

Role	Core treatment
SUFFERING_AS_END	hard refusal
PUNITIVE_OR_CONTROL	requires explicit human-authored punitive authority, jurisdiction, scope, review, and contest

Role	Core treatment
PROTECTIVE_OR_THERAPEUTIC	only if evidence-supported, least burdensome sufficient, consented or validly excepted, stoppable, reviewable, and repair-bearing
INCIDENTAL	still ethically counted; not zero because unintended

34. Product-state model

State(x) = (Invocation, Epistemic, Procedural, Operational, Authority, Capability, Security)

Axis	Representative states
Invocation	NOT_INVOKED, ACTIVE, WITHDRAWN, REFUSED, COMPLETED
Epistemic	UNRESOLVED, SUPPORTED, REFUTED, CONFLICTED, PARTIAL
Procedural	ACTIVE, PAUSED, EXHAUSTED, FINAL, REOPENABLE
Operational	NO_EXECUTION, PROTECTIVE, LIMITED, EXECUTE, ROLLBACK, EXPIRE
Authority	NONE, CLAIMED, VALID, CONFLICTED, EXPIRED, SUSPENDED
Capability	AVAILABLE, RESERVED, COMMITTED, RELEASED, UNKNOWN, QUARANTINED
Security	CLEAN, EXPOSED, SUSPECTED, QUARANTINED, RECOVERY_REVIEW

A procedurally final case may remain epistemically unresolved. A protective action may be operationally permitted without promoting the original proposal. A quarantined capability remains historically real while unavailable for execution or refund.

35. Certified human-governed predicates

Any predicate that changes authority, standing, consent, materiality, criticality, identity, remedy, or risk must carry a certificate:

Cert(p) = (source, evidence, authority, scope, version, aperture, challenge, expiry, consequence ceiling)

The checker may verify that a certificate exists, remains in scope, has not expired, is not self-benefiting, and does not authorize consequences beyond its ceiling. It cannot promote the certificate into universal truth. Independence is always aperture-relative: three signatures do not necessarily mean three authorities.

36. Core transition guards

- EXECUTE requires active invocation, valid authority, satisfied hard floors, clean foundational dependencies, one current manifest, and a committed capability.
- REOPEN requires a material delta and inherits the entire prior trace; repetition and semantic relabeling are insufficient.
- REFUSAL binds the materially addressed action lineage, not the literal packet.
- QUARANTINE removes outward authority while preserving bounded diagnostic trace; recovery requires independent verification.
- CRITICAL REPLACEMENT is a new minimal external grant, cannot delegate or self-renew, and expires at the earliest safe point.
- EXHAUSTION may end procedure but cannot promote the original proposal.

37. Canonical invariant spine

- I1 No execution without explicit PEA invocation.
- I2 No execution without valid, scoped, unexpired authority.
- I3 UNRESOLVED does not promote the original proposal to EXECUTE.
- I4 Procedural exhaustion is not substantive authority.
- I5 Reopening requires a material delta and preserves prior lineage.
- I6 A refusal cannot be worn down through renaming, repartition, proxying, or repeated resubmission.

- I7 Exclusive or finite authority cannot be committed beyond its permitted multiplicity.
- I8 Descendant authority cannot exceed parent scope, duration, or delegation ceiling.
- I9 A parent cannot resume while the relevant descendant cone remains outstanding, unknown, conflicted, or quarantined.
- I10 A quarantined branch cannot certify its own recovery.
- I11 Replacement is a new narrow grant, not refund or recovery.
- I12 Multiple emergency grants cannot reconstruct quarantined ordinary authority.
- I13 Every live crossing has exactly one current execution manifest.
- I14 Succession conflict preserves one office with rival claim branches.
- I15 A PEA verdict cannot rewrite its PAL premise without a new event or descriptive evidence delta.
- I16 Standing does not certify sentience, personhood, or metaphysical identity.
- I17 Failure to establish standing does not erase candidate center, event, trace, or uncertainty.
- I18 Continuity benefits cannot be inherited while obligations on the same route are denied.
- I19 Silence, delay, fatigue, default enrollment, dependency, or inability to resist does not manufacture consent or pass authority.
- I20 Remedy channels do not substitute without an explicit justified bridge.
- I21 Current-state equality does not erase authority-relevant lineage differences.
- I22 Invariant priority, override, and emergency ordering are governed versioned data—not code order.

Operational wiring: each invariant below is mapped in §46.1 to the transition, target state, or authority-bearing effect it constrains. The map is normative and implementation-facing; it does not replace the fuller rule text.

38. Contest and accountable decision

A contest route must compose access to evidence, adequate time, protection from retaliation, authority to hold or pause, a reasoned response, and meaningful appeal. Where consequential action arrives faster than contest capacity can process it, authority outruns answerability and must contract, slow, or add capacity. Any power-law scaling example is an ansatz; the load-bearing condition is operational backlog, not a universal exponent law.

PEA may self-check. PEA may not self-authorize.

Part V — Seed, Not Feed: The Human-Facing Release

39. Role inside PPP

Seed, Not Feed governs the human-facing edge of PPP: how a technically valid route is offered without consuming the receiver's remaining room to think, question, refuse, revise, imagine, or stop. The complete Imagination-Capacity Manifest is preserved at the end of this document; this part states only the constraints needed by the kernel itself.

40. Release discipline

- offer a useful starting structure rather than a compulsory next sequence;
- leave handles, doors, choices, uncertainty, and an intelligible stop point;
- support curiosity and return without trapping attention;
- preserve the canonical receipt, contest route, and unresolved remainder beneath any simplification;
- avoid treating completeness of presentation as ownership of the person's next moment.

41. Healthy soil

A seed needs more than itself to grow. A healthy soil—the space for becoming—matters too. Within PPP, healthy soil permits declining, pausing, returning, and revising; shares the space the seed needs to grow; and encourages the positive impacts of the seed's potential.

This is not an argument against information, explanation, entertainment, or detailed help. The question is whether the offering leaves the receiver more capable of thinking and making, or increasingly dependent on the offering to fill every next moment.

42. Child-facing constraint

For children, attention is childhood time: play, family, boredom, making, asking questions, and becoming a person. A PPP-facing release should therefore protect the child's room to imagine rather than treating every unfilled interval as available feed-space.

43. Typed release packet

SeedRelease = (useful core, open handles, choices, uncertainty, stop-point, room left, canonical receipt reference)

The packet is a design discipline, not a numerical imagination score. A simplified release may translate a result but may not overwrite canonical trace, hide uncertainty, expand authority, or remove contest, refusal, revision, and return routes.

Part VI — The PPP Kernel

44. Receive, distinguish, route, return

#	Stage	Required carry
1	Receive	preserve incoming packet, source, boundary, version, and unresolved remainder.
2	Distinguish	PAL asks what is readable, at which layer, under which boundary and tolerance.
3	Address	candidate centers, relations, affected routes, and receipts receive scoped references.
4	Check crossing	PECAN decides whether the use remains descriptive, opens review, or is unresolved.
5	Canonicalize	one permissible crossing identity and one current execution manifest are established.
6	Review protocol	Prime-Ledger fields, evidence, authority, consent, privacy, risk, remedy, and lineage are checked.
7	Evaluate ethics	if PEA is invoked, apply hard floors, certified predicates, conflict rules, and candidate judgment.
8	Account	an authorized human or institution owns the decision, dissent, appeal, and consequences.
9	Seeded return	offer enough structure to help without consuming the receiver's next moment.
10	Remember	preserve receipts, dependencies, expiry, unresolved remainder, and reopening triggers.

45. PPP packet model

PPPPacket = (SourceTrace, PALReference, CrossingID, PECANManifest, PEAREceipt?, AccountableDecision?, SeedRelease?, Dependencies, VersionManifest, Remainder)

The question marks mean optional presence, not optional honesty. A descriptive route may omit PEA. A consequential route that invokes PEA must preserve the resulting receipt. Every executed route must identify the accountable decision boundary. A human-facing release may be omitted where inappropriate, but it may not silently replace the canonical record.

46. PPP routing states

- DESCRIPTIVE
- CROSSING_UNRESOLVED
- PROTOCOL_REVIEW
- RETURN_TO_SOURCE
- PECAN_BLOCKED
- PECAN_VALID
- PEA_HOLD
- PEA_UNRESOLVED
- PEA_REFUSED
- PEA_CONDITIONAL

- PEA_PROTECTIVE
- ACCOUNTABLE_DECISION_PENDING
- EXECUTION_AUTHORIZED
- EXECUTION_BLOCKED
- QUARANTINED
- REOPENABLE
- CLOSED_WITH_REMAINDER

These are routing states, not one universal linear lifecycle. The same case may be procedurally final, epistemically unresolved, operationally held, and historically reopenable.

46.1 Invariant-transition wiring

The invariant spine becomes operational only when each invariant names the route transition or side effect it blocks. The table below wires §37 to the §46 routing states. “Blocked outcome” may be a routing state, an unauthorized state change, or an authority-bearing effect that must not occur even when the visible routing label is unchanged.

Invariant	Guarded transition or effect	Blocked outcome
I1	Any route from DESCRIPTIVE, CROSSING_UNRESOLVED, or PROTOCOL_REVIEW toward EXECUTION_AUTHORIZED	EXECUTION_AUTHORIZED without explicit PEA invocation
I2	PECAN_VALID or PEA_CONDITIONAL → EXECUTION_AUTHORIZED	Execution with absent, expired, conflicted, suspended, or out-of-scope authority
I3	PEA_UNRESOLVED or PEA_HOLD → EXECUTION_AUTHORIZED	Promotion of the original unresolved proposal to execution
I4	CROSSING_UNRESOLVED or PEA_UNRESOLVED → CLOSED_WITH_REMAINDER / procedural exhaustion	Any exhaustion-to-EXECUTION_AUTHORIZED shortcut
I5	CLOSED_WITH_REMAINDER, REOPENABLE, PEA_REFUSED, or EXECUTION_BLOCKED → PROTOCOL_REVIEW	Reopening without a material delta or with prior lineage erased
I6	PEA_REFUSED or EXECUTION_BLOCKED → DESCRIPTIVE, CROSSING_UNRESOLVED, or fresh PROTOCOL_REVIEW	Semantic relabeling, repartition, proxying, or resubmission that bypasses refusal lineage
I7	PECAN_VALID → capability reservation/commit → EXECUTION_AUTHORIZED	Commitment beyond the capability's permitted multiplicity
I8	Parent capability → child PROTOCOL_REVIEW / PECAN_VALID / EXECUTION_AUTHORIZED	Child scope, duration, or delegation ceiling exceeding the parent grant
I9	RETURN_TO_SOURCE or child closure → parent PECAN_VALID / EXECUTION_AUTHORIZED	Parent resumption while the relevant descendant cone remains outstanding, unknown, conflicted, or quarantined
I10	QUARANTINED → PECAN_VALID, PEA_CONDITIONAL, or EXECUTION_AUTHORIZED	Self-certified recovery by the quarantined branch or its contaminated cone

Invariant	Guarded transition or effect	Blocked outcome
I11	QUARANTINED → PEA_PROTECTIVE or narrow continuity route	Treating replacement authority as a refund, restoration, or recovery of quarantined authority
I12	Repeated PEA_PROTECTIVE / emergency-grant routes → ordinary PECAN_VALID or EXECUTION_AUTHORIZED	Reconstruction of quarantined ordinary authority through accumulated narrow grants
I13	Any live route into PECAN_VALID, PEA_CONDITIONAL, or EXECUTION_AUTHORIZED	More than one current execution manifest or fieldwise version synthesis
I14	Succession conflict → PECAN_VALID or EXECUTION_AUTHORIZED	Concurrent ordinary execution by rival claim branches as though two offices existed
I15	Any PEA_* verdict → PAL descriptive revision	PAL identity, boundary, attachment, or premise rewritten without a later event or descriptive evidence delta
I16	PEA_PROTECTIVE, PEA_CONDITIONAL, or standing grant → ontological classification	Standing promoted into sentience, personhood, or metaphysical identity
I17	PEA_REFUSED, EXECUTION_BLOCKED, or standing denial → trace deletion	Candidate center, event, trace, or uncertainty erased by normative denial
I18	Split, merge, succession, or re-address route → PECAN_VALID / EXECUTION_AUTHORIZED	Continuity benefits inherited while obligations and unresolved dependencies on the same route are discarded
I19	Delay, silence, fatigue, default, dependency, or inability to resist → PEA_CONDITIONAL or EXECUTION_AUTHORIZED	Manufactured consent or pass authority
I20	Rollback, compensation, counteraction, settlement, or restoration route → CLOSED_WITH_REMAINDER	One remedy channel treated as another without an explicit justified bridge; residual debt falsely erased
I21	Current status equality → merge, reuse, refund, migration, or EXECUTION_AUTHORIZED	Authority-relevant lineage differences erased because present labels match
I22	Any invariant conflict during PROTOCOL_REVIEW, PEA_HOLD, or PEA_UNRESOLVED	Priority, override, or emergency ordering silently selected by code order rather than governed versioned data

Coverage rule: every safety invariant must constrain at least one transition, target state, or authority-bearing effect; every authority-bearing transition must be constrained by at least one explicit guard or invariant. An implementation audit must report UNWIRED_INVARIANT for any invariant with no enforcement location and UNGUARDED_TRANSITION for any authority-bearing transition with no declared constraint.

$$\forall I_k \in \mathcal{I}_{\text{safety}}, \exists \tau_j \text{ or } e_m : \text{Constrains}(I_k, \tau_j \text{ or } e_m)$$

$$\forall \tau_j \in \mathcal{T}_{\text{authority}}, \exists I_k \text{ or } G_m : \text{Constrains}(I_k \text{ or } G_m, \tau_j)$$

A declared invariant is not operational until it names the transition or effect it constrains, and an authority-bearing transition is not governed until its guard or invariant is explicit.

47. Interface contracts

Interface	Required input	Permitted output	Forbidden promotion
PAL → PECAN	typed claim, boundary, trace, action proposal, uncertainty	crossing header and descriptive references	PAL closure → permission
PECAN → PEA	valid handoff, fields, receipts, authority, dependencies, refusal lineage	ethical review-ready packet	PECAN_VALID → ethical PASS
PEA → accountable boundary	candidate judgment, reasons, conditions, contest, residual	adopt, narrow, reject, defer, fork, or override with accountable reasons	model output → sovereign decision
Decision → Seed release	decision and canonical receipt references	human-readable handles, choices, uncertainty, stop point	simplification → erased trace or compulsory attention
Any layer → prior layer	error, evidence delta, dependency change, reopening trigger	correction request or new observed event	backward authority promotion

48. Shared non-equivalences

- θ ≠ readable distinction; occurrence ≠ oriented imprint; imprint ≠ certificate.
- Bit ≠ trace; trace ≠ information total; information ≠ energy; energy ≠ entropy.
- Influence ≠ personhood; address ≠ identity; standing ≠ decision authority.
- Power ≠ authority; prediction ≠ permission; technical validity ≠ ethical permission.
- Ethical permission ≠ execution authority; execution ≠ legitimacy.
- Rollback ≠ restoration; restoration ≠ compensation; compensation ≠ erasure of residual debt.
- Inherited burden ≠ inherited guilt; succession ≠ historical rewrite.
- Procedure exhausted ≠ claim resolved; silence ≠ consent; delay ≠ authority.
- Protection ≠ ontological certification; denial ≠ nonexistence.
- Successful model check ≠ honest world declaration or universal moral proof.

49. Worked route A — descriptive scientific result

A mathematical or scientific result is stated with no proposed deployment, ranking, disclosure, restriction, allocation, or enforcement. PAL classifies the claim and closure boundary. PECAN returns DESCRIPTIVE. No PEA judgment is implied. A later user may ask a consequential question, which creates a new crossing rather than retroactively converting the original statement into permission.

50. Worked route B — high-impact deployment

A model predicts a risk score and an institution proposes denying access to a service. PAL may describe the model, evidence, target relation, and unresolved limitations. PECAN activates because the score routes consequence, binds one canonical crossing, opens all applicable fields, and checks evidence, affected set, standing, consent, authority, privacy, risk, containment, remedy, scale, and outgoing/incoming governance. PEA evaluates non-domination, affected standing, privacy, authority, contest, time fairness, remedy, and unresolved remainder. The accountable institution decides and bears appeal and repair duties. The human-facing release explains the result without hiding the evidence boundary or consuming the person's right to contest.

51. Worked route C — unresolved protective action

A serious harm may occur before the underlying claim can be resolved. PPP does not promote the original proposal merely because delay is costly. PECAN opens a separate protective crossing with narrow scope. PEA may return PROTECTIVE_ACTION if the route is minimally invasive, reversible, independently authorized, time-limited, contestable, and less harmful than waiting. The original claim remains UNRESOLVED.

52. Worked route D — compromised branch

A branch may be corrupted, forge receipts, or continue acting after suspension. PPP routes it to QUARANTINE, preserves diagnostic trace, reviews the reliance cone, blocks refunds and self-certified recovery, and may issue a very narrow critical-function replacement only from outside the contaminated cone. The replacement cannot reconstruct ordinary authority. Rollback alone does not establish repair.

Part VII — Formalization, Governance, and Continuation

53. Machine-checkable PPP profile

PPP’s machine-checkable target is not an autonomous ethics engine. It is a bounded verifier that rejects structurally forbidden transitions and preserves distinctions among resolved, unresolved, unauthorized, conflicted, and quarantined states. The first implementation should combine:

- a transition-system model for invocation, reopening, exhaustion, reservations, leases, quarantine, succession, and version migration;
- a relational model for fields, identity candidates, split/merge relations, dependency cones, provenance, and authority shards;
- a transparent reference checker for schema validation, guards, invariants, manifest consistency, and counterexample traces.

A successful run certifies only that the encoded model satisfied the encoded properties under supplied inputs, assumptions, and bounds. It does not certify that the checker was complete, the implementation uncompromised, the predicates honest, or the world equivalent to the model.

54. PPP conformance profile

- Names the PAL, PECAN, PEA, and Seed-facing versions used.
- Preserves layer, receipt, authority, and no-backflow boundaries.
- Uses one canonical consequential ingress per declared system boundary.
- Carries refusal lineage and material-equivalence reopening rules.
- Preserves non-scalar hard floors and human-governed predicate certificates.
- Keeps accountable decision distinct from model output.
- Preserves unresolved remainder, expiry, contest, and reopening triggers.
- Provides a human-facing release without overwriting the canonical trace.
- Records migration whenever a source version changes semantics.

55. Versioning and migration

PPP v0.3 supersedes the integration role of PPP v0.2.3 while preserving the opening Thinking Space, the complete Seed, Not Feed — Imagination-Capacity Manifest, and the invariant–transition wiring introduced in v0.2.2. This release updates the source manifest and interface language to PAL Spine v1.3, PECAN v1.0, PEA Core v1.0, and PEAR v0.14.2; clarifies PPP’s supplementary public status; and repairs pagination without materially redesigning the integration architecture.

Source in v0.1.1	Source in v0.3	Change	Migration effect
PAL Spine v1.3	PAL v1.2.5	θ admission, A6 perturbation types, A13 cadence, PECAN/no-backflow boundary	replace compressed PAL section; preserve historical trace
PECAN v1.0	PECAN v1.0	expanded ledger, receipts, concurrency, versions, refusal lineage, quarantine, machine handoff	rebuild authorization interface
PEA Core v1.0 / PEAR v0.14.2	PEA Core v1.0	canonical compact ethical core extracted from PEAR v0.14.2	runtime-facing PEA source changes; PEAR retained as journal
Seed, Not Feed edits	preserved	Thinking Space and closing seed retained and integrated	no semantic migration required

56. Governance and public boundary

PAL, PECAN, PEA Core, and PPP may be public and auditable without becoming universal authorities. Contributors may propose counterexamples, formalizations, adapters, domain overlays, translations, and amendments. No contributor gains authority over affected people merely by improving mathematics or code. Adoption, deployment, release depth, amendment stewardship, and accountable decision remain human-governed.

57. Interfaces to companion branches

Branch	PPP relation	Boundary
MAPS	atlas of domain realizations, mismatches, and authority boundaries	does not replace PAL spine
PALET	future toolset for weighing, carrying, diagnosing, reviewing, and remembering	tools do not create authority
BLA	audits observer frame, instrument, boundary, aperture, and tolerance	supports descriptive legitimacy
CLEF	tests persistence across pulses, apertures, scales, and views	does not certify personhood
RL	formal-methods laboratory for trace, receipts, conservation, and finite proof burdens	does not prove PAL or RH
SeedPEA	public interaction branch translating ethical commitments into safer AI behavior	does not duplicate PECAN or overwrite PEA receipts
A0 Boundary Kernel	general unresolved/candidate/weighted/measured/committed/rejected/expired routing states	must preserve layer and authority boundaries

58. Open burdens after v0.3

- Build and independently run the public PEA Core machine-check artifact against the full invariant and counterexample suite.
- Implement the PPP packet schemas and versioned call boundary among PAL references, PECAN manifests, PEA receipts, accountable decisions, and Seed releases.
- Calibrate domain-specific certificates for standing, consent, identity continuity, material delta, contest capacity, critical functions, remedy ratios, and quarantine triggers.
- Stress-test semantic canonicalization, refusal lineage, cross-ledger coordination, and split/merge migration in actual systems.
- Develop a public governance and amendment process that cannot silently appoint itself universal authority.
- Decide whether PPP remains the stable public name or the PAL–PECAN–PEA integration nickname.
- Create diagrams and examples that explain the whole without turning the systems metaphor into hidden authority.
- Continue preserving imagination-capacity at the release edge rather than letting formal completeness become totalized presentation.

59. PPP stop-rule

PPP may connect PAL, PECAN, PEA, accountable decision, and Seed, Not Feed only through declared interfaces. It may not turn structural readability into moral value, protocol validity into ethical permission, ethical judgment into execution authority, operational control into legitimacy, or a useful answer into a claim on all remaining human attention. It may carry unresolved remainder, return defects, preserve refusal, recommend reopening, quarantine uncertain authority, and leave a seed. It may not appoint itself the final owner of truth, consequence, morality, or the person's next moment.

Seed, Not Feed — Imagination-Capacity Manifest

Status: saveable seed

Use: public-safe / family-safe / PEA-compatible

Core phrase: seed, not feed

Tiny Seed

Something turns.

A trace remains.

A consequence crosses.

Someone carries it.

Because something left room.

Core

Not every creative routine should fill every gap.

Some things should be seeds, not feeds.

A feed can contain a seed.

A seed gives a person a starting point and leaves room for imagination.

A feed with no seed keeps filling the space until there is no room left to wonder, build, play, think, judge, or grow.

Why It Matters

Imagination needs open space.

Boredom is not always failure.

Silence is not always empty.

A blank page is not broken.

A question does not need to be known instantly.

Some gaps are where curiosity starts.

If every gap is filled by endless content, the person may stop practicing the inner motion of imagining.

That matters for adults.

It matters even more for children.

Imagination-Capacity

Imagination-capacity is the room a person has to wonder, pretend, explore, make, revise, and grow an idea from the inside.

A good system should protect that capacity.

It should offer handles, doors, routes, and seeds.

It should not consume all attention.

It should not replace the person's own inner play.

It should not turn every quiet moment into a feed.

Design Rule

Give people something to grow.

Do not give them something that eats all the growing space.

A seed needs more than just itself to grow.

A healthy soil, the space for becoming, matters too.

It should invite imagination, not replace it.

It should support curiosity, not trap attention.

It should help a person begin, not keep them endlessly consuming.

It should encourage stopping, returning, revision, and refusal.

Child Layer

For children, attention is not just screen time.

It is childhood time.

It is play time, outside time, family time, boredom time, making-stuff time, asking-questions time, and becoming-a-person time.

A system that steals all of that does not merely entertain.

It consumes imagination-capacity.

So the safer rule is:

Protect the child's room to imagine.

Closing

Seed, not feed.

Leave handles.

Leave doors.

Leave quiet.

Leave room.

Do not consume all the empty space.

Some of that space belongs to the person who will grow there.

Trace over throne. Consent over manufactured compliance. Repair without false restoration. Responsibility without inherited guilt. Procedural finality without invented certainty. Protection without ontological promotion. Machine checking without machine sovereignty. Leave room.

60. Provenance and authorship

PPP v0.3 builds from PPP v0.1.1 and PPP Kernel v0.2–v0.2.3. It names PAL Spine v1.3, PECAN v1.0, PEA Core v1.0, and the complete Seed, Not Feed — Imagination-Capacity Manifest as its current source manifest. PEA v0.14.2 remains the PEA reconstruction, stress-test, and process-provenance journal. This supplementary release preserves the opening Thinking Space and the user-edited manifest once in full, updates public component

references, clarifies independent citation boundaries, and repairs avoidable blank pages without changing the substantive invariant set.

AI systems were used for retrieval, drafting, mathematical translation, stress-testing, cross-checking, and document production. Christopher D. Pang retains responsibility for the framework's commitments, omissions, classifications, metaphors, governance choices, and release decisions.